
Algorithm 1 Pseudo code of BASO

Require: Population size N , Max iterations T , Parameters α, β , Dataset with F features.

Ensure: Optimal binary feature subset X_{gb} .

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1: Randomly initialize binary population  $X$  and velocities  $V$ 
2: Evaluate fitness for all atoms and find global best  $X_{gb}$  and  $Fit_{gb}$ 
3: for  $t = 1$  to  $T$  do
4:   Calculate mass  $M_i$  for all atoms using Eq. (2)
5:   Determine subset of best atoms  $K_{best}$  using Eq. (3)
6:   Update time-varying parameters  $\eta$  and  $\lambda$  using Eq. (4) and (5)
7:   for  $i = 1$  to  $N$  do
8:     Calculate length scale  $\sigma_i$  relative to centroid of  $K_{best}$  atoms (Eq. 6)
9:     for  $j \in K_{best}$  do
10:      Calculate Euclidean distance  $r_{ij}$ 
11:      Calculate normalized distance  $h_{ij}$ 
12:      Compute Lennard-Jones force magnitude  $F_{mag}$ 
13:      Generate random weight  $r_j$ 
14:    end for
15:    Calculate total acceleration  $A_i$  using Eq. (9)
16:    for  $d = 1$  to  $F$  do
17:      Update velocity:  $V_i^d = \text{clip}(\text{rand}(0, 1) \cdot V_i^d + A_i^d, -V_{max}, V_{max})$ 
18:      Calculate selection probability:  $S = f_{\text{TruePRLU}}(V_i^d)$ 
19:      if  $\text{rand}(0, 1) < S$  then
20:         $X_i^d \leftarrow 1$ 
21:      else
22:         $X_i^d \leftarrow 0$ 
23:      end if
24:    end for
25:  end for
26:  Evaluate fitness for all atoms
27:  if new best fitness found then
28:    Update  $X_{gb}$  and  $Fit_{gb}$ 
29:  end if
30: end for
31: return  $X_{gb}$ 
```
